Vertical organisation of microbial communities in Salineta hypersaline wetland, Spain

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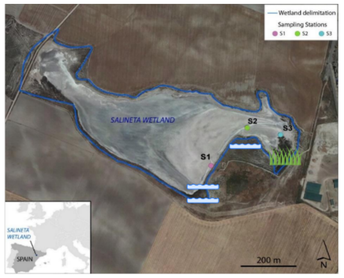
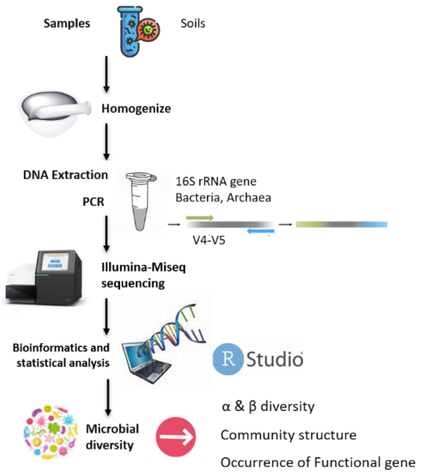
INTRODUCTION

Saline lakes are numerous and geographically widespread in inland ecosystems. The Salineta at "Saladas de Bujaraloz-Sástago" (Spain), located in south of Monegros, one of the most arid areas in Europe, is an endorheic complex constituted by a large set of inland ephemeral saline lakes subjected to temporary flooding according to seasons.
Salineta wetland, surrounded by new irrigated areas, is characterized by three different landscapes corresponding to water flooding conditions: usually submerged soil, intermittently flooded soil, and soil being vegetated with halophytes.



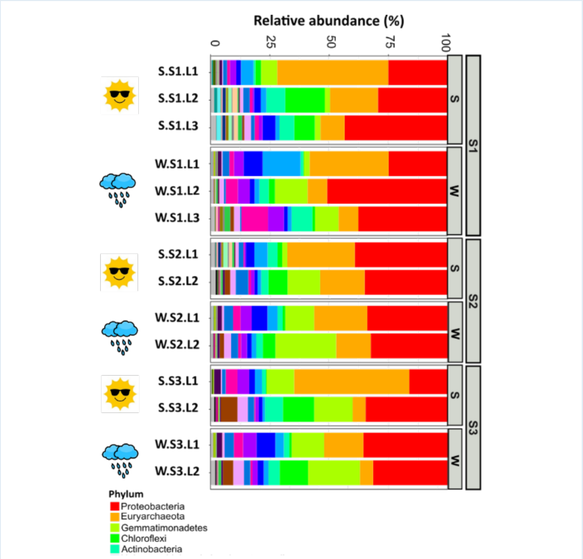
Location of the sampling stations (S1, S2 and S3) at Salineta wetland

METHODS

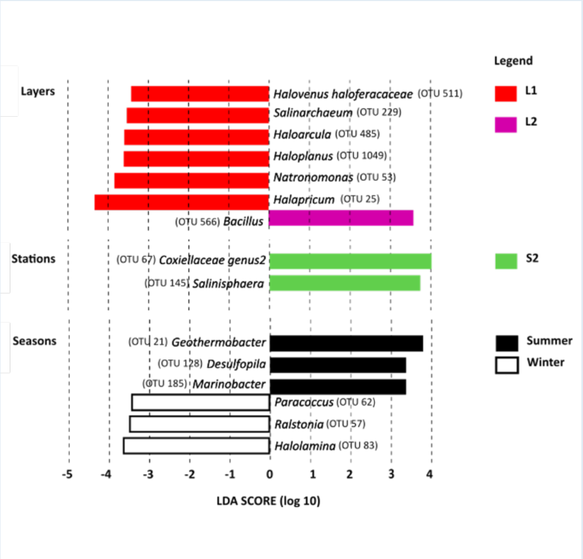


Physical and chemical parameters: Salinity, redox, ions, OM, Total nitrogen ...

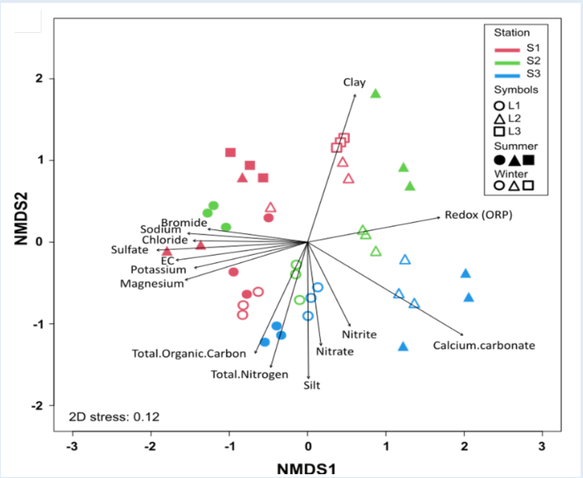
RESULTS



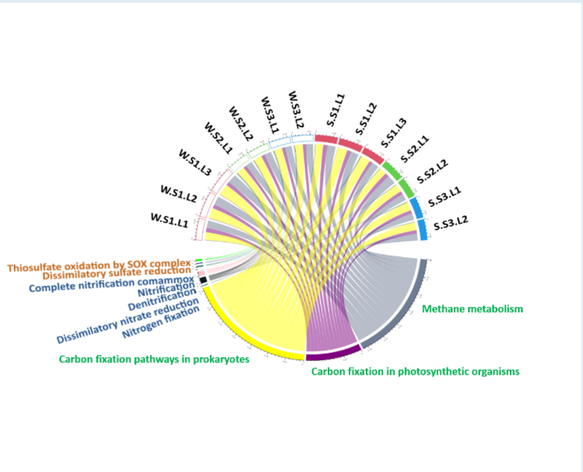
The microbial community was composed by 78% of Bacteria and 22 % of Archaea. The microbial communities were affected by the conditions prevailing in the stations according to layers and seasons.



Linear discriminant effect size analysis revealed OTUs found significantly more abundant according to layers, stations, and seasons, which serve as biomarkers. The layer L1 was characterised by archaeal OTU biomarkers, exclusively halophilic archaea. Regarding stations biomarkers, only two OTUs were identified for S2, these OTUs represent taxa well adapted to salinity fluctuating conditions to intermittent flooding. Seasonal biomarkers related to the salinity variations according to wet and dry periods.

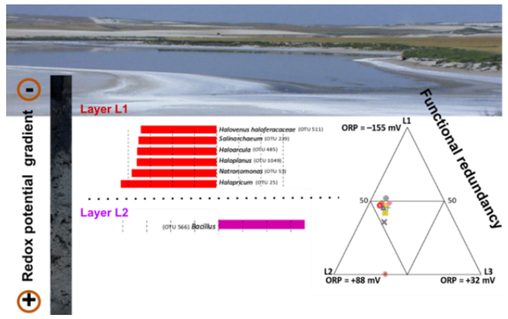


The comparison of microbial communities separated the layers in clusters irrespective of the stations and seasons indicating that the variation between samples was explained mainly by layer. The microbial community organization was driven by environmental factors, L1 being controlled by ion concentrations, organic matter and total nitrogen, whereas L2 mainly influenced by redox.



The functional capabilities involved in the main biogeochemical cycles (nitrogen, sulfur and carbon) were determined. Focusing on nitrogen, sulfur, and carbon metabolisms, the functional profiles were similar in all samples indicating that same metabolic functions are ensured by different microbial communities.

CONCLUSION



The three sampling stations in Salineta wetland within a gradient of water presence were all characterized by reversed redox gradient in the soil profile. Thus, the soil layers were the main drivers explaining microbial diversity distribution. Although they exhibited different taxonomic microbial communities, the predictive functional analysis revealed similar functions involved in the main biogeochemical cycles (nitrogen, sulfur and carbon) in all the soil layers suggesting functional redundancy. The functioning in this highly dynamic ecosystem is ensured by the redundancy of the biogeochemical cycle functions.

ACKNOWLEDGEMENTS

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